

Andrés Sáurez Apuy

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Machine learning researcher working on the geometry of transformer representations: why linear interpretability methods work, and what that implies for reading and controlling model behaviour. Six publications, two first-author, across language models, vision-language models, and reinforcement learning, alongside a computer vision system deployed at a live industrial site. M.S. from KAIST; before that, three years as a software engineer across backend, frontend, and QA.

EDUCATION

KAIST | M.S., Robotics Program (Artificial Intelligence & Machine Learning) Sep 2023 – Feb 2026
Thesis: *Invariant Feature Geometry in Transformer Representations*. Advisor: Prof. Dongsoo Har.

Universidad de Costa Rica | B.S., Mechanical Engineering Mar 2014 – Dec 2019
Outstanding Thesis Award.

RESEARCH EXPERIENCE

Graduate Researcher | VIL Lab, KAIST | Advisor: Prof. Dongsoo Har Sep 2023 – Feb 2026

Identity-Projection: Invariant Subspaces from Architectural Constraints (first author, under review at NeurIPS 2026)

- Proved that linear unembedding makes every logit margin a projection of the residual stream onto a fixed contrast direction, while nonlinear readouts remove this necessity, isolating linear structure as a property of the readout interface rather than of cross-entropy. Confirmed by a grokking ablation: Fourier embeddings emerge under linear and two-layer readouts and vanish under deeper nonlinear ones, as predicted.
- Derived the Self-Reference Property from a sparse-superposition projection-error bound: a class prompt's own activation serves as a label-free reference for its feature direction, enabling zero-shot classification in a single forward pass. SAE overlap analysis confirms class-token and instance activations recruit the same dictionary atoms.
- Validated across 8 classification tasks, 4 model families (Llama3-8B, Mistral-7B, GPT-2 Small, Llama3.2-3B), and 3 architectures (decoder-only LLMs, BERT, ViT/LLaVA). Extended to circuit attribution: the projection-based variant outperforms AtP and AtP* on IOI in GPT-2 Small, with predicted and observed direct logit attribution matching at slope 1.07, $R^2 = 0.92$.

Circuit Fingerprints: How Answer Tokens Encode Their Geometrical Path (first author, preprint)

- Showed that answer tokens encode their own computational pathways, enabling gradient-free circuit discovery from geometric alignment at performance comparable to gradient-based methods for a fraction of the compute. Evaluated on IOI, SVA, and MCQA with CPR/CMD metrics across GPT-2, Qwen2.5, Llama 3.2, and OPT.
- Showed that circuit discovery and activation steering are dual operations on the same geometric structure, supporting the reading of discovered circuits as causal mechanisms rather than correlational artefacts.
- Extended to feature circuits via prompt engineering: geometric steering reaches 69.8% emotion-classification accuracy against 53.1% for instruction prompting, while holding factual accuracy at 89.6% (91.2% baseline).

Visual grounding in large VLMs (second author, TMLR 2026, J2C Certification)

- Applied attention-head analysis and direct logit attribution to frozen LLaVA, showing that a small number of contrastive grounding heads localise referred objects, which enables training-free referring image segmentation with no labelled data or fine-tuning.

Autonomous EV Charging Robot — Industry Collaboration, VIL Lab, KAIST

- Built and deployed the computer vision pipeline for a 6-DoF robotic arm performing autonomous charge-port detection and connector insertion, running in an EV parking tower at the Korea Transportation Safety Authority's Advanced Automotive Inspection Research Center (Gimcheon).
- Two-stage detection (YOLO localisation, then classical CV for rotation and centre refinement) with two-pass measurement to suppress stereo-camera noise on low-cost hardware. Python, OpenCV, YOLO.

PUBLICATIONS

- Under review, **NeurIPS 2026** — *Why Linear Interpretability Works: Invariant Subspaces from Architectural Constraints*. First author.
- Preprint (arXiv) — *Circuit Fingerprints: How Answer Tokens Encode Their Geometrical Path*. First author.
- **TMLR 2026** — *Few Contrastive Attention Heads Enable Visual Grounding in Large VLMs*. Second author. J2C Certification (~10% of accepted TMLR papers).
- Under review, **IEEE TCSVT** — *CAGSeg: Training-Free Referring Image Segmentation via Contrastive Attribution Grounding in Vision-Language Models*. Second author.
- **IJCAI 2026** — *DEPLOY-RL: Active Boundary Discovery for Deployable RL in Safety-Critical Processes*.
- **ICAII 2025** — *Continuous Adversarial Text Representation Learning for Affective Recognition*. Second author. Excellent Paper Award.

ENGINEERING EXPERIENCE

Backend Software Engineer | TransUnion Apr 2023 – Aug 2023

- Sole engineer on the end-to-end migration of an IP-based geo-localisation service from monolith to microservices: owned architecture, API design, service boundaries, implementation, and rollout, under high-availability constraints and without disrupting downstream consumers. Java, Spring, REST, Git.

Software Engineer | ValueLabs

Feb 2021 – Apr 2023

- Led the modernisation of a company-wide Java Swing desktop application: built foundational classes reused across the project and served as the team's primary contact for Swing-related technical issues.
- QA engineering on the Krames/WebMD web app: automated regression suites in Selenium, resolved bugs blocking sprint releases, and ran manual and automated test cycles. Java, Java Swing, C#, Selenium, JavaScript, MySQL, GitLab.

Frontend Software Engineer | Upwork

Aug 2022 – Aug 2023

- Implemented changes across Vue.js pages and components to support multi-language internationalisation of the Upwork platform; refactored Vue/Nuxt pages to improve page load time, time-to-interactive, and error handling. JavaScript, TypeScript, Vue.js, Nuxt, PHP.

TECHNICAL SKILLS

Interpretability: TransformerLens, sparse autoencoders, linear probing, activation patching and ablation, direct logit attribution, circuit discovery, activation steering, attention-head analysis

LLMs: PyTorch, Hugging Face Transformers, Llama 3.2, Qwen2.5, Mistral, GPT-2, OPT, BERT; prompt engineering, fine-tuning, evaluation-set design, benchmarking

ML / AI: Deep learning, transformers, computer vision (OpenCV, YOLO), reinforcement learning; NumPy, Pandas, scikit-learn

Maths: Linear algebra and projection geometry, probability, optimisation, statistical validation

Engineering: Python, Java, TypeScript / JavaScript, C++, SQL; Spring, microservices, REST APIs, Docker, Git, Linux, CI workflows

ADDITIONAL

Awards: Excellent Paper Award (ICAIIC 2025); J2C Certification (TMLR 2026); Outstanding Thesis Award (Universidad de Costa Rica)

Certifications: AWS Certified Cloud Practitioner

Languages: Spanish (native), English (TOEFL 104), Korean (lower-intermediate)